

Data

This chapter describes the datasets used throughout the book, one possibility is that this could be moved to an appendix and in the online version of the book any mention of the dataset in the main text could be cross-linked to its description. One thing that I really liked in the first edition is the emphasis on analysis question of interest and the little notes about the data. I believe the original layout was to put the data in order of appearance of the book, which naturally clusters them by technique.

Name	Description	Source
aflw	Player statistics from the AFLW	cassowaryr
bushfires	Multivariate spatio-temporal data for locations of bushfires	spotaroo
oz_election_2001	Socioeconomic characteristics of Australian electorates in 2001	https://github.com/jf
oz_election_2016	Socioeconomic characteristics of Australian electorates in 2016	https://github.com/jf
penguins	Measure four physical characteristics of three species of penguins	https://allisonhorst.g
pisa	OECD programme for international student assessment data	learningtower
sketches	Google's Quickdraw data	https://quickdraw.wit
clusterchallenge	Simulated data used to show various cluster examples	liminal

Australian Football League Women

Description

The `aflw` data is from the 2021 Women's Australian Football League. These are average player statistics across the season, with game statistics provided by the [fitzRoy](#) package. If you are new to the game of AFL, there is a nice explanation on [Wikipedia](#).

Variables

Rows: 381

Columns: 35

```
$ id          <chr> "CD_I1001678", "CD_I1001679", "CD_I1001681", "CD_I1001~
$ given_name  <chr> "Jordan", "Brianna", "Jodie", "Ebony", "Emma", "Pepa",~
$ surname     <chr> "Zanchetta", "Green", "Hicks", "Antonio", "King", "Ran~
$ number      <int> 2, 3, 5, 12, 60, 21, 22, 23, 35, 14, 3, 8, 16, 12, 19,~
$ team        <chr> "Brisbane Lions", "West Coast Eagles", "GWS Giants", "~
$ position    <chr> "INT", "INT", "HFFR", "WL", "RK", "BPL", "INT", "INT",~
$ time_pct    <dbl> 63.00000, 61.25000, 76.50000, 74.90000, 85.10000, 77.4~
$ goals       <dbl> 0.0000000, 0.0000000, 0.0000000, 0.1000000, 0.6000000,~
$ behinds     <dbl> 0.0000000, 0.0000000, 0.5000000, 0.4000000, 0.4000000,~
```

```

$ kicks          <dbl> 5.000000, 2.500000, 3.750000, 8.800000, 4.100000, 3.22~
$ handballs      <dbl> 2.500000, 3.750000, 3.000000, 3.600000, 2.700000, 2.22~
$ disposals      <dbl> 7.500000, 6.250000, 6.750000, 12.400000, 6.800000, 5.4~
$ marks          <dbl> 1.500000, 0.250000, 1.000000, 3.700000, 2.200000,~
$ bounces        <dbl> 0.000000, 0.000000, 0.000000, 0.600000, 0.100000,~
$ tackles        <dbl> 3.000000, 2.250000, 2.250000, 3.900000, 2.000000, 1.77~
$ contested      <dbl> 3.500000, 2.250000, 3.500000, 5.700000, 4.400000, 2.66~
$ uncontested    <dbl> 3.500000, 4.500000, 3.000000, 7.000000, 2.800000, 1.77~
$ possessions    <dbl> 7.000000, 6.750000, 6.500000, 12.700000, 7.200000, 4.4~
$ marks_in50     <dbl> 1.000000, 0.000000, 0.250000, 0.500000, 0.900000,~
$ contested_marks <dbl> 1.000000, 0.000000, 0.000000, 0.400000, 1.200000,~
$ hitouts        <dbl> 0.000000, 0.000000, 0.000000, 0.000000, 19.400000~
$ one_pct        <dbl> 0.000000, 1.500000, 0.500000, 1.200000, 2.600000,~
$ disposal       <dbl> 60.25000, 67.15000, 37.20000, 65.96000, 61.72000, 66.8~
$ clangers       <dbl> 2.000000, 0.500000, 2.500000, 3.100000, 2.400000, 1.33~
$ frees_for      <dbl> 1.000000, 0.500000, 0.250000, 2.500000, 0.500000,~
$ frees_against  <dbl> 1.000000, 0.500000, 1.250000, 1.300000, 1.100000,~
$ rebounds_in50  <dbl> 0.000000, 0.500000, 0.250000, 1.100000, 0.000000,~
$ assists        <dbl> 0.000000, 0.000000, 0.000000, 0.200000, 0.2000~
$ accuracy       <dbl> 0.00000, 0.00000, 0.00000, 5.00000, 30.00000, 0.00000,~
$ turnovers      <dbl> 1.500000, 1.000000, 2.500000, 4.000000, 1.700000, 1.22~
$ intercepts     <dbl> 2.000000, 2.000000, 0.500000, 5.300000, 1.300000,~
$ tackles_in50   <dbl> 0.500000, 0.000000, 0.750000, 0.500000, 0.500000,~
$ shots          <dbl> 0.500000, 0.000000, 0.750000, 1.000000, 1.200000,~
$ metres         <dbl> 72.50000, 58.50000, 76.00000, 225.90000, 89.80000, 76.~
$ clearances     <dbl> 0.500000, 0.250000, 1.250000, 0.400000, 0.900000,~

```

Purpose

The primary analysis is to summarise the variation using principal component analysis, which gives information about relationships between the statistics or skills sets common in players. One also might be tempted to cluster the players, but there are no obvious clusters so it could be frustrating. At best one could partition the players into groups, while recognising there are no absolutely distinct and separated groups.

Source

See the information provided with the [fitzRoy](#) package.

Pre-processing

The code for downloading and pre-processing the data is available at the [mulgar](#) website in the `data-raw` folder. The data provided by the `fitzRoy` package was pre-processed to reduce the variables to only those that relate to player skills and performance. It is possible that using some transformations on the variables would be useful to make them less skewed.

Bushfires

Description

This data was collated by Weihao (Patrick) Li as part of his Honours research at Monash University. It contains fire ignitions as detected from satellite hotspots, and processed using the [spotaroo](#) package, augmented with measurements on weather, vegetation, proximity to human activity. The cause variable is predicted based on historical fire ignition data collected by County Fire Authority personnel.

Variables

Rows: 1,021

Columns: 60

```
$ id      <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 1~
$ lon     <dbl> 141.1300, 141.3000, 141.4800, 147.1600, 148.1050, 144.18~
$ lat     <dbl> -37.13000, -37.65000, -37.35000, -37.85000, -37.57999, -~
$ time    <date> 2019-10-01, 2019-10-01, 2019-10-02, 2019-10-02, 2019-10~
$ FOR_CODE <dbl> 41, 41, 91, 44, 0, 44, 0, 102, 0, 91, 45, 41, 45, 45, 45~
$ FOR_TYPE <chr> "Eucalypt Medium Woodland", "Eucalypt Medium Woodland", ~
$ FOR_CAT  <chr> "Native forest", "Native forest", "Commercial plantation~
$ COVER    <dbl> 1, 1, 4, 2, 6, 2, 6, 5, 6, 4, 2, 1, 2, 2, 2, 2, 6, 6, 6,~
$ HEIGHT   <dbl> 2, 2, 4, 2, 6, 2, 6, 5, 6, 4, 3, 2, 3, 3, 3, 2, 6, 6, 6,~
$ FOREST   <dbl> 1, 1, 1, 1, 0, 1, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0,~
$ rf       <dbl> 0.0, 0.0, 15.4, 4.8, 6.0, 11.6, 11.6, 0.6, 0.2, 0.6, 0.0~
$ arf7     <dbl> 5.0857143, 2.4000000, 2.4000000, 0.7142857, 0.8571429, 1~
$ arf14    <dbl> 2.8142857, 1.7428571, 1.8000000, 1.6714286, 1.5714286, 1~
$ arf28    <dbl> 1.9785714, 1.5357143, 1.5357143, 3.7857143, 1.9000000, 1~
$ arf60    <dbl> 2.3033333, 1.7966667, 1.7966667, 4.0000000, 2.5333333, 1~
$ arf90    <dbl> 1.2566667, 1.0150000, 1.0150000, 2.9600000, 2.1783333, 1~
$ arf180   <dbl> 0.9355556, 0.8444444, 0.8444444, 2.3588889, 1.7866667, 1~
$ arf360   <dbl> 1.3644444, 1.5255556, 1.5255556, 1.7272222, 1.4716667, 1~
$ arf720   <dbl> 1.3011111, 1.5213889, 1.5213889, 1.7111111, 1.5394444, 1~
$ se       <dbl> 3.8, 4.6, 14.2, 23.7, 23.8, 16.8, 18.0, 12.9, 14.7, 12.9~
```

\$ ase7	<dbl> 18.02857, 18.50000, 21.41429, 23.08571, 23.11429, 22.014~
\$ ase14	<dbl> 17.03571, 17.44286, 18.03571, 19.17143, 18.45714, 18.628~
\$ ase28	<dbl> 19.32857, 18.47500, 19.33929, 18.23571, 16.86071, 19.375~
\$ ase60	<dbl> 20.38644, 19.99153, 20.39492, 19.90847, 19.26780, 20.449~
\$ ase90	<dbl> 22.54118, 21.93193, 22.04370, 20.59328, 20.04538, 21.809~
\$ ase180	<dbl> 20.79106, 19.93966, 19.99385, 19.11006, 18.66760, 19.810~
\$ ase360	<dbl> 15.55153, 14.83259, 14.87883, 14.69276, 14.44318, 14.755~
\$ ase720	<dbl> 15.52350, 14.75049, 14.77427, 14.53463, 14.32656, 14.540~
\$ maxt	<dbl> 21.3, 17.8, 15.4, 20.8, 19.8, 15.8, 19.5, 12.6, 18.8, 12~
\$ amaxt7	<dbl> 22.38571, 20.44286, 22.21429, 24.21429, 23.14286, 21.671~
\$ amaxt14	<dbl> 21.42857, 19.72857, 19.86429, 21.80000, 20.89286, 19.578~
\$ amaxt28	<dbl> 20.71071, 19.10000, 19.18929, 19.75000, 19.05714, 18.885~
\$ amaxt60	<dbl> 24.02667, 22.28000, 22.38667, 22.93167, 22.12000, 21.031~
\$ amaxt90	<dbl> 27.07750, 25.77667, 25.89833, 24.93667, 23.93750, 23.164~
\$ amaxt180	<dbl> 26.92000, 25.92722, 25.98500, 24.84056, 23.95389, 23.343~
\$ amaxt360	<dbl> 21.55389, 20.79778, 20.81333, 20.21972, 19.99389, 19.505~
\$ amaxt720	<dbl> 21.47750, 20.57222, 20.57694, 20.13153, 20.03875, 19.650~
\$ mint	<dbl> 9.6, 9.0, 7.3, 7.7, 8.3, 8.3, 6.1, 5.9, 7.4, 5.9, 6.9, 7~
\$ amint7	<dbl> 9.042857, 7.971429, 9.171429, 10.328571, 11.200000, 10.6~
\$ amint14	<dbl> 9.928571, 9.235714, 9.421429, 10.007143, 10.900000, 10.7~
\$ amint28	<dbl> 8.417857, 7.560714, 7.353571, 8.671429, 9.575000, 10.060~
\$ amint60	<dbl> 11.156667, 9.903333, 9.971667, 10.971667, 11.975000, 12.~
\$ amint90	<dbl> 11.96667, 10.81250, 10.87833, 12.49000, 13.46167, 13.638~
\$ amint180	<dbl> 11.96778, 11.01056, 11.02000, 12.41944, 13.42500, 13.695~
\$ amint360	<dbl> 9.130556, 8.459722, 8.448333, 9.588611, 10.456389, 11.03~
\$ amint720	<dbl> 8.854861, 8.266250, 8.254028, 9.674861, 10.517083, 10.96~
\$ dist_cfa	<dbl> 9442.206, 6322.438, 7957.374, 7790.785, 10692.055, 6054.~
\$ dist_camp	<dbl> 50966.485, 6592.893, 31767.235, 8816.272, 15339.702, 941~
\$ ws	<dbl> 1.263783, 1.263783, 1.456564, 5.424445, 4.219751, 4.1769~
\$ aws_m0	<dbl> 2.644795, 2.644795, 2.644795, 5.008369, 3.947659, 5.2316~
\$ aws_m1	<dbl> 2.559202, 2.559202, 2.559202, 5.229680, 4.027398, 4.9704~
\$ aws_m3	<dbl> 2.446211, 2.446211, 2.446211, 5.386005, 3.708622, 5.3045~
\$ aws_m6	<dbl> 2.144843, 2.144843, 2.144843, 5.132617, 3.389890, 5.0355~
\$ aws_m12	<dbl> 2.545008, 2.545008, 2.548953, 5.045297, 3.698736, 5.2341~
\$ aws_m24	<dbl> 2.580671, 2.580671, 2.584047, 5.081100, 3.745286, 5.2522~
\$ dist_road	<dbl> 498.75145, 102.22032, 1217.22446, 281.69151, 215.56176, ~
\$ log_dist_cfa	<dbl> 9.152945, 8.751860, 8.981854, 8.960697, 9.277256, 8.7084~
\$ log_dist_camp	<dbl> 10.838924, 8.793748, 10.366191, 9.084354, 9.638200, 9.15~
\$ log_dist_road	<dbl> 6.212108, 4.627130, 7.104329, 5.640813, 5.373247, 5.0047~
\$ cause	<chr> "lightning", "lightning", "lightning", "lightning", "lig~

Purpose

The primary goal is to predict the cause of the bushfire using the weather and distance from human activity variables provided.

Source

Collated data was part of Weihao Li's Honours thesis, which is not publicly available. The hotspots data was collected from @jaxa, climate data was taken from the Australian Bureau of Meteorology using the **bomrang** package [@R-bomrang], wind data from @wind and @windasos, vegetation data from @forest, distance from roads calculated using @OpenStreetMap, CFA stations from @cfa, and campsites from @recreation. The cause was predicted from training data provided by @fireorigin.

Pre-processing

The 60 variables are too many to view with a tour, so it should be pre-processed using principal component analysis. The categorical variables of FOR_TYPE and FOR_CAT are removed. It would be possible to keep these if they are converted to dummy (binary variables).

Australian election data

Description

This is data from a study on the relationship between voting patterns and socio-demographic characteristics of Australian electorates reported in @eechidna. These are the predictor variables upon which voting percentages are modelled. There are two years of data in oz_election_2001 and oz_election_2016.

Variables

```
load("data/oz_election_2001.rda")
load("data/oz_election_2016.rda")
glimpse(oz_election_2001)
```

Purpose

The tour is used to check for multicollinearity between predictors, that might adversely affect the linear model fit.

Source

The data was compiled from Australian Electoral Commission (AEC) and the Australian 38 Bureau of Statistics (ABS). Code to construct the data, and the original data are available at <https://github.com/jforbes14/eechidna-paper>.

Pre-processing

Considerable pre-processing was done to produce these data sets. The original data was wrangled into tidy form, some variables were log transformed to reduce skewness, and a subset of variables was chosen.

Palmer penguins

```
library(palmerpenguins)
penguins <- penguins %>%
  na.omit() # 11 observations out of 344 removed
# use only vars of interest, and standardise
# them for easier interpretation
penguins_sub <- penguins %>%
  select(bill_length_mm,
         bill_depth_mm,
         flipper_length_mm,
         body_mass_g,
         species,
         sex) %>%
  mutate(across(where(is.numeric), ~ scale(.)[,1])) %>%
  rename(bl = bill_length_mm,
         bd = bill_depth_mm,
         fl = flipper_length_mm,
         bm = body_mass_g)
save(penguins_sub, file="data/penguins_sub.rda")
```

Description

This data measure four physical characteristics of three species of penguins.

Variables

Name	Description
bl	a number denoting bill length (millimeters)
bd	a number denoting bill depth (millimeters)
fl	an integer denoting flipper length (millimeters)
bm	an integer denoting body mass (grams)
species	a factor denoting penguin species (Adélie, Chinstrap and Gentoo)

Purpose

The primary goal is to find a combination of the four variables where the three species are distinct. This is also a useful data set to illustrate cluster analysis.

Source

Details of the penguins data can be found at <https://allisonhorst.github.io/palmerpenguins/>, and @R-palmerpenguins is the package source.

Pre-processing

The data is loaded from the `palmerpenguins` package. The four physical measurement variables and the species are selected, and the penguins with missing values are removed. Variables are standardised, and their names are shortened.

```
library(palmerpenguins)
penguins <- penguins %>%
  na.omit() # 11 observations out of 344 removed
# use only vars of interest, and standardise
# them for easier interpretation
penguins_sub <- penguins[,c(3:6, 1)] %>%
  mutate(across(where(is.numeric), ~ scale(.)[,1])) %>%
  rename(bl = bill_length_mm,
         bd = bill_depth_mm,
```

```

    fl = flipper_length_mm,
    bm = body_mass_g) %>%
  as.data.frame()
save(penguins_sub, file="data/penguins_sub.rda")

```

Program for International Student Assessment

Description

The `pisa` data contains plausible scores for math, reading and science of Australian and Indonesian students from the 2018 testing cycle. The plausible scores are simulated from a model fitted to the original data, to preserve privacy of the students.

Variables

Name	Description
CNT	country, either AUS for Australia or IDN for Indonesia
PV1MATH-PV10MATH	plausible scores for math
PV1READ-PV10READ	plausible scores for reading
PV1SCIE-PV10SCIE	plausible scores for science

Purpose

Primarily this data is useful as an example for dimension reduction.

Source

The full data is available from <https://www.oecd.org/pisa/>. There are records of the student test scores, along with survey data from the students, their households and their schools.

Pre-processing

The data was reduced to country and the plausible scores, and filtered to the two countries. It may be helpful to know that the SPSS format data was used, and was read into R using the `read_sav()` function in the `haven` package.

Sketches

Description

This data is a subset of images from <https://quickdraw.withgoogle.com>. The subset was created using the quickdraw R package at <https://huizezhang-sherry.github.io/quickdraw/>. It has 6 different groups: banana, boomerang, cactus, flip flops, kangaroo. Each image is 28x28 pixels. The `sketches_train` data would be used to train a classification model, and the unlabelled `sketches_test` can be used for prediction.

Variables

Name	Description
V1-V784	grey scale 0-255
word	what the person was asked to draw, NA in the test data
id	unique id for each sketch

Purpose

Primarily this data is useful as an example for supervised classification, and also dimension reduction.

Source

The full data is available from <https://quickdraw.withgoogle.com>.

Pre-processing

It is typically useful to pre-process this data into principal components. This code can also be useful for plotting one of the sketches in a recognisable form:

```
library(mulgar)
library(ggplot2)
data("sketches_train")
set.seed(77)
x <- sketches_train[sample(1:nrow(sketches_train), 1), ]
xm <- data.frame(gry=t(as.matrix(x[,1:784])),
                 x=rep(1:28, 28),
```

```

      y=rep(28:1, rep(28, 28)))
ggplot(xm, aes(x=x, y=y, fill=gry)) +
  geom_tile() +
  scale_fill_gradientn(colors = gray.colors(256, start = 0, end = 1, rev = TRUE )) +
  ggtitle(x$word) +
  theme_void() +
  theme(legend.position="none")

```

crab



Figure 1: One of the sketches in the subset of training data.

Links to additional data used in the book

	Chapter	Name	Link
	4	afw tour path	https://dicook.github.io/mulgar_boo
	7	detourr saved clusters	https://dicook.github.io/mulgar_boo
	7	detourr saved clusters	https://dicook.github.io/mulgar_boo
	1	2001 election	https://dicook.github.io/mulgar_boo
	1	2016 election	https://dicook.github.io/mulgar_boo
5, 7, 8, 9, 10, 11, 12, 13, 14, 15		Penguins re-organsed	https://dicook.github.io/mulgar_boo
8, 10, 13, 14		2D tour path for penguins	https://dicook.github.io/mulgar_boo
	12	risk survey	https://dicook.github.io/mulgar_boo